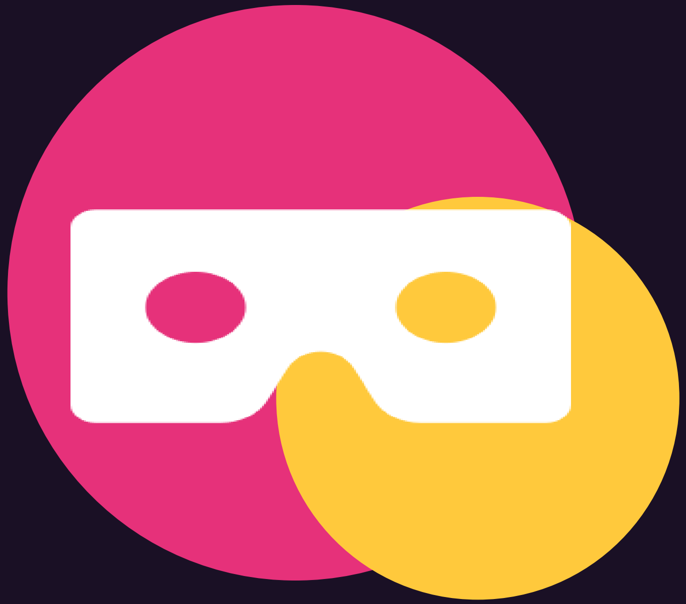
SEMINAR TOPIC NO. 27

CSA1016



Augmented Reality (AR) & Virtual Reality (VR)

Blurring the boundary between the physical and digital worlds

Presented by Rashmi Naurene(192521357)

Guided By: Dr. K. Anita Davamani

INTRODUCTION

What Is It & Why It Matters

What Is AR/VR?

- AR overlays digital content on the real world (phone camera, smart glasses).
- VR replaces reality with a fully immersive digital environment.
- Together they form the spectrum of Extended Reality (XR).

Why It Matters

- Powers immersive learning, training & simulation.
- Enables remote collaboration and telepresence.
- Fastest-growing frontier in gaming, retail & design.
- Lets users rehearse risky tasks safely.

AR Example-Pokemon Go

AR Example-Snapchat filters

VR Example-Google expeditions

VR Example-VR Games

ARCHITECTURE

How AR/VR Works: Sense → Process → Render → Interact

1 SENSE

Cameras, IMUs & depth sensors capture the real world and user motion.

2 PROCESS

SLAM tracking & AI fuse sensor data to build a live spatial map.

3 RENDER

The graphics engine draws the 3D scene in real time, frame by frame.

4 INTERACT

Controllers & haptics let the user touch, move and respond naturally.

APPLICATIONS

Real-World Impact

Gaming & Entertainment

Healthcare & Therapy

Education & Training

Retail & Manufacturing

Advantages

- Deeply immersive, hands-on learning experiences.
- Cuts travel, prototyping & training costs.

Limitations

- High hardware cost & motion discomfort for some users.
- Heavy compute & bandwidth demands.

LOOKING AHEAD

Trends, Challenges & the Road Ahead

RECENT TRENDS

- AI-generated 3D worlds & avatars
- Lightweight standalone headsets
- Spatial computing (Apple Vision Pro)
- 5G/6G-enabled cloud rendering

CHALLENGES AHEAD

- Device cost & comfort limit mass adoption.
- Motion sickness and eye strain for extended use.
- Privacy of spatial & biometric data.
- Scarcity of high-quality immersive content.

Case Study: IKEA Place lets shoppers preview true-to-scale furniture in their own room via AR — boosting confidence and cutting returns.

AR/VR is moving computing from the screen into the space around us — the next interface is everywhere.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